

EE204 Semiconductor Devices

HW 5

1. A Si p-n junction has donor and acceptor doping concentrations N_d and N_a of 10^{16} cm^{-3} and 10^{18} cm^{-3} , respectively. Assume $kT = 0.026 \text{ eV}$ at 300 K, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_r = 11.8$. Given that the junction has an area of 10^{-4} cm^2 , calculate the junction capacitance at (i) zero bias (ii) FB of 0.65 V and (ii) RB of 2 V.
(3.18 pF, 7 pF, 1.7pF)
2. Room temperature (300 K) measurements on a silicon p^+ -n junction diode yield the following results: With a reverse bias of 4.3 V, the junction capacitance is 20 pF. When the reverse bias is changed to 0.55 V, the junction capacitance is 40 pF. If $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, $\epsilon_r = 11.8$, and the diode area is $2 \times 10^{-3} \text{ cm}^2$, determine the contact potential V_o , and the doping density of the n-side N_d .
(0.7 V, $6 \times 10^{15} \text{ cm}^{-3}$)
3. Consider a silicon p^+ -n junction with $N_d = 10^{16} \text{ cm}^{-3}$. The area of the diode is 10^{-4} cm^2 . Assume $kT = 0.026 \text{ eV}$ at 300 K, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_r = 11.8$. Holes in the n-region have a lifetime of 0.5 μs with a diffusion coefficient of $10 \text{ cm}^2/\text{s}$. Calculate the storage capacitance of the p-n junction at a forward bias of 0.65 V.
(2.23 nF)
4. An ideal Schottky contact is formed on n-type Si having a donor concentration of 10^{17} cm^{-3} . The bandgap of Si at 300 K is 1.1 eV. The metal work function ϕ_m is 5.4 eV and the electron affinity χ is 4.5 eV at 300K. Calculate the semiconductor work function ϕ_s . Also draw the energy band diagrams (a) before and after contact formation at equilibrium (b) at a forward bias of 0.2 V and (c) reverse bias voltage of 3.2 V. Assume $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.
(4.642 eV)
5. Draw the energy band diagrams of a Schottky contact before and after the formation of contact at the thermal equilibrium based on metal and n-type semiconductor. Repeat the energy band diagrams of the contact at forward and reverse bias, clearly showing the changes in the energy levels due to biases.
6. A metal semiconductor junction is formed between an n-doped Si and aluminum. It is given that the Si has a doping concentration $N_d = 10^{17} \text{ cm}^{-3}$ and a work function

$q\phi_s = 4.191$ eV, and aluminum has a work function $q\phi_m = 4.3$ eV. Assume that $kT = 0.026$ eV, and Si has a bandgap of 1.1 eV and intrinsic concentration $n_i = 1.5 \times 10^{10}$ cm⁻³. Deduce the electron affinity $q\chi$ of Si. Draw the energy band diagrams before and after the junction formation under equilibrium condition. Hence determine whether the junction formed is rectifying or ohmic. All the relevant energies must be clearly indicated in the band diagrams.

(4.05 eV, Schottky contact)

7. Discuss the operation of Ohmic contacts based on energy band diagrams.